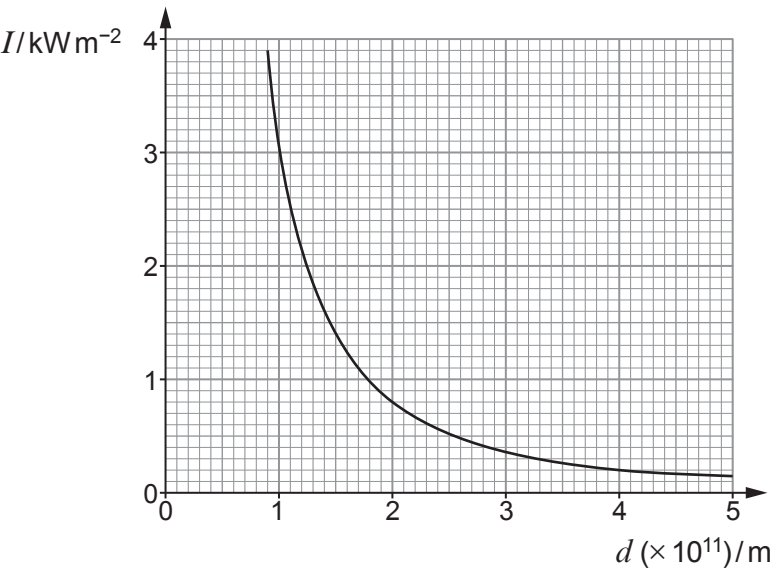


7. (a) Stars are very good approximations of *black bodies*. Explain what is meant by the term *black body*. [1]

.....

.....

(b) The graph shows how the intensity of electromagnetic radiation, I , from the Sun varies with the distance from its centre, d .



(i) Confirm that the graph shows an inverse square relationship between I and d . [2]

.....

.....

.....

.....

.....

(ii) Show that the Sun's luminosity is approximately 4×10^{26} W. [2]

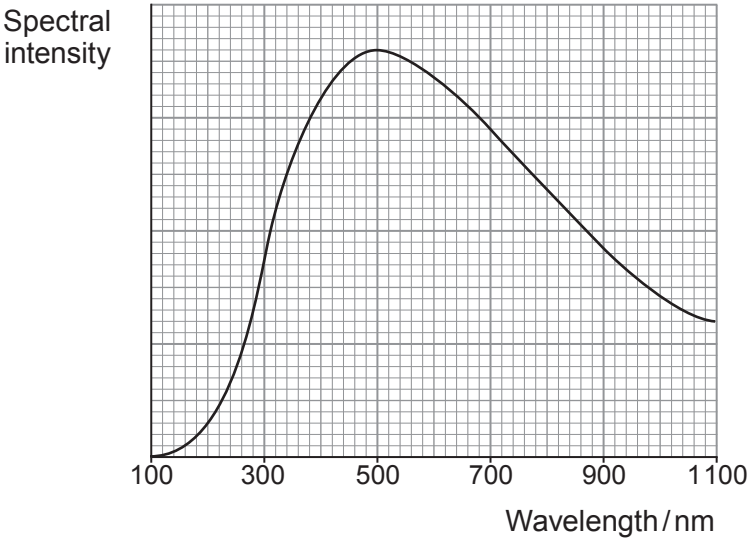
.....

.....

.....



- (c) The graph shows how the intensity of the radiation incident on the Earth from the Sun is distributed across the spectrum.



Use information from this graph, along with your answer to (b)(ii), to calculate the radius of the Sun. [4]

.....

.....

.....

.....

.....

.....

END OF PAPER

